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New early Jurassic paleopole from France and Jurassic apparent polar wander

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ABSTRACT

A new early Jurassic paleopole for stable Europe obtained from a doleritic dyke system in Brittany, France is given at latitude 61°N and longitude 79°E ($K = 36$ and $A_{95} = 10.2°$) based on seven equivalent sites and more than a hundred samples demagnetized by alternating field and thermal treatments ($D = 41°$, $I = 62°$, $k = 66$, $\alpha_{95} = 7.5°$). Radiometric datings provide ages between 190 and 200 Ma. This result represents the first indication obtained from European igneous data for a very westerly cusp in the Apparent Polar Wander Path around 200 Ma. The European data also tend to support the hypothesis of a higher latitude track for the Jurassic APWP as already proposed from North American data.

1. Introduction

The study of the paleomagnetism of the Kerforne dolerite dyke system in Brittany has spanned two decades. In the 1970's results from a preliminary study [1] were in agreement with some poles from England [2–4] but not with the APWP defined for Europe. They were therefore interpreted as representing a short excursion of the APWP or reflecting the existence of a large non-dipole field [1], and they were therefore rejected from subsequent compilations [e.g. 5] as being unrepresentative of the stable lower Jurassic paleopole for Europe. In the 1980's, as part of a paleomagnetic study of the Jurassic rocks of France, two sites were added to those previously studied along the Kerforne dyke system and new K-Ar ages were obtained [6–8], and in the late 1980's further sampling was conducted for paleointensity studies [9]. Some results from this last study confirm the primary character of the Kerforne magnetization. We summarize here all the results obtained from the Kerforne dykes.

2. Geological setting and sampling

A subvertical dyke system cuts the Hercynian and pre-Hercynian structures of the western extremity of the Massif Armoricain over a distance of more than 70 km along the NW–SE striking (N165°E) Kerforne fault system (Fig. 1). This tholeiitic magmatism [10] was related to the very early opening of the North Atlantic Ocean and was active by the end of the Hercynian phase. With such a dense dyke swarm [13], the thickness of the individual dykes ranges from a few centimetres to some tens of metres, and therefore only the major dykes and veins are indicated on the map (Fig. 1).

The Kerforne dyke system has been the subject of many K-Ar dating studies which have yielded the following dates: 197 ± 10 Ma (whole rock) [11], 194 ± 10 Ma (whole rock) and 189 ± 10 Ma (feldspar) [12], 208 ± 10 Ma (whole rock) [13], 191 ± 10 Ma (whole rock) [14] and 155 ± 8 Ma (whole rock) [15]. This discrepancy of the latter date led Bellon and Fabre to conduct a new K-Ar study on whole rock samples and on mineral separates [7,8]. Two groups of ages, with no intermediate values between the two end-member ages, were obtained from the Brenterc'h samples: the older age (200 ± 8 Ma, Early Jurassic) was interpreted as the minimum age of the intrusion, the youngest age (156 ± 8 Ma, Middle Jurassic) as the result of hydrothermal alteration (seritiza-

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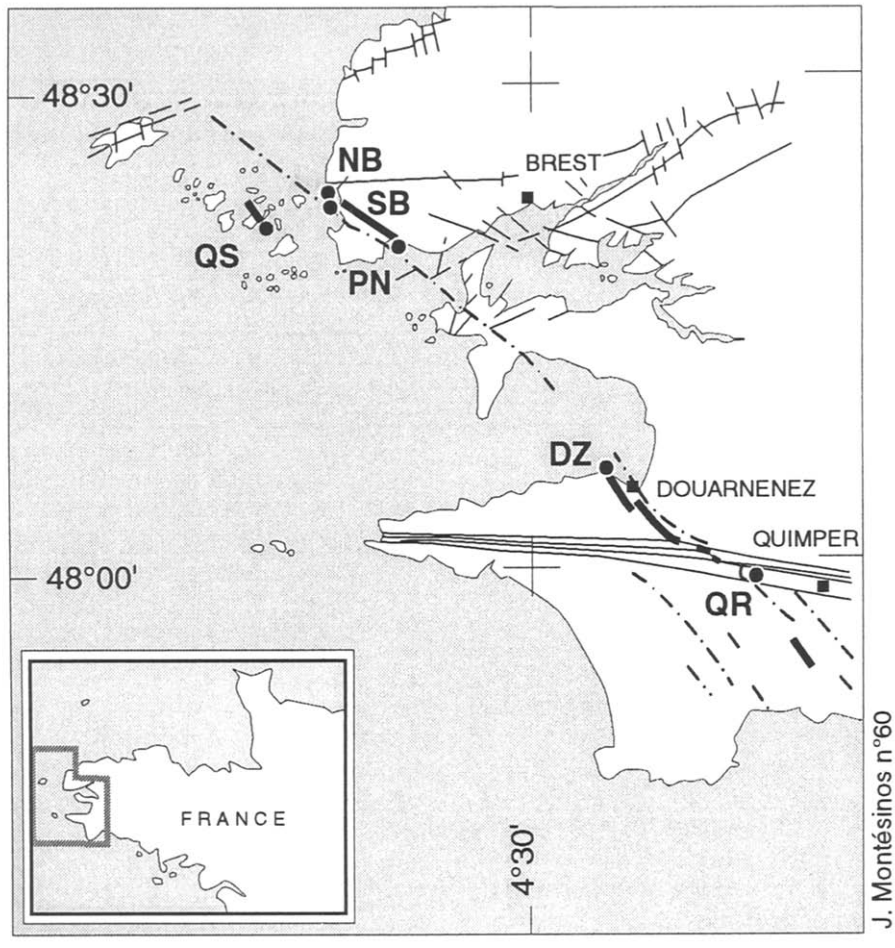


Fig. 1. Outline of the Kerforne dyke system in Brittany, northwestern France, and sampling sites. *QS* = Quemenes; *NB* and SB = north and south Brenterc'h; *PN* = Porsmilin; *DZ* = Douarnenez; *QR* = Quimper. Heavy lines = Kerforne dyke system; dashed lines = Kerforne fault system.

tion). Finally, a recent $^{40}Ar/^{39}Ar$ determination on plagioclases gave an age of 190 ± 2 Ma [16].

All our samples were drilled out directly in the field with a petrol-powered core drill. Orientation of the samples was carried out using both magnetic and sun compasses. During the first trips, 56 samples were collected at six different sites. All sites but one were situated in outcrops along

TABLE 1

Sampling information

Locality	Lat °N	Long °E	First sampling		Second sampling	
			N	Site	*N*	Site
Quemenes	48.38	355.11	12	QS1		
North Brenterc'h	48.39	355.23	7	NB1	27	NB2
South Brenterc'h	48.39	355.23	13	SB1	42	SB2
Porsmilin	48.35	355.32	8	PN1		
Douarnenez	48.10	355.64	8	DZ1		
Quimper	48.01	355.83	8	QR1		

the shore (Table 1 and Fig. 1). To define lower Jurassic paleointensity, the Brenterc'h site was visited again and 69 very fresh, fine- to medium-grained samples were collected at two outcrops (Table 1 and Fig. 1). Between two and seven specimens were obtained from each sample.

The extreme northwestern part of the Kerforne dyke system (i.e. the Brenterc'h site) is bifurcated, with a thickness of the order of 30 m for the northern branch (site NB) and of 15 m for the southern branch (site SB). The two sites are easily accessible at low tide. In thin section, there is no petrographic difference between the two branches. The samples were of quartz tholeiites with a doleritic, poecilitic structure. The outcrop of the dykes is embedded in the Ploumoguer gneiss and has been well scoured by the waves, so the dolerites are very fresh. Following the dyke, towards the southeast, the next site was established at the shoreline at Porsmilin (site PN), in the Brest gneiss. Its position in the Bay of Brest screens it from the heavy swell of the open sea and consequently the dolerites are more weathered. The Douarnenez site (DZ) occurs in the same physical geographical setting as the others, and its dykes are embedded the Locronan alkali leucogranites. The only inland site is located near Quimper (site QR), in a granite quarry. The final site was sampled on the Quemenes islet (site QS), where a narrow (4 m) doleritic dyke outcrops subparallel to the Kerforne fault, along which the dyke is shifted by 5 km. Veinlets with calcite, chlorite, epidote and clay minerals are evidence for severe low-temperature weathering. Whole rock K-Ar dating yields ages that are greater than those of the other sites: 242 ± 12 Ma [17] and 233 ± 12 Ma [8].

3. Tectonic setting

Geological studies conducted in the Western Approaches have demonstrated two post-Paleozoic tectonic phases, the major one occurring between the Paleocene and the Eocene (60 Ma) and the minor one some 50 Ma ago, between the Ypresian and Lutetian [18].

North of the dyke system, a FLEXOTIR seismic profile, which was shot by SNPA between Ouessant and the Scilly Isles [19,20], displays, near Ouessant, horizontal layers that have been ascribed to the upper Cretaceous; these layers lie upon a Permo-Triassic basement. Reflection and refraction profiles shot during the GEOMARGE cruise [21] show horizontal Mesozoic and Paleogene layers south of the studied zone near Ouessant; beyond the 'Ouessant trough' to the north, these layers are 'rippled', gently warped, and disturbed by faults.

To the south the dyke system occurs under a small sedimentary basin in Concarneau Bay, in which an upper Lutetian layer shows an insignificant tilt of 1° to the east [22].

These two sedimentary controls at the tips of the dyke system and the vertical selvedges along 70 km of its length provide an important tectonic constraint in that they have prevented any general or differential tilt. Nevertheless, some horizontal and vertical shifts (without rotations) cannot be eliminated where there is no sedimentary control. According to Lefort [20] a possible kinematic model for the Kerforne fault system would be a succession of vertical and shear movements since the Paleozoic.

4. Laboratory experiments and magnetic properties

Samples from the first sampling (Table 1) were studied at the Ifremer Paleomagnetic Laboratory in Brest, whereas the second set (Table 1) was studied at the U.S. Geological Survey Paleomagnetic Laboratory at Menlo Park in the USA. Remanent magnetizations were measured with a spinner magnetometer at Ifremer and a cryogenic magnetometer in Menlo Park. Initial magnetic susceptibility (χ) at room temperature was measured with a kappa bridge for NB1 samples and with a 1000 Hz frequency commercial bridge for the second set of samples. Most of the alternating field (AF) experiments were carried out at Ifremer on the first sampling. Stepwise AF demagnetization was carried out from 5 to 50 mT at steps of 5–10 mT. Stepwise thermal treatments were also carried out on this set of samples, from 250 to 600°C using 50–100° steps. For the second set, one specimen of each sample having a viscosity index less than 5% (one of the selection criteria for paleointensity analysis [9]) was thermally demagnetized with through six to eight steps, from 100–200°C to 550–575°C.

In both cases, the magnetic viscosity was first estimated by Thellier's procedure [24]. Viscosity indexes ($v\%$) defined from the Brenterc'h and Porsmilin sites are fairly low ($< 10\%$), whereas those calculated from the Quemenes, Douarnenez and Quimper sites are much higher (22%, $30 \pm 4\%$ and $123 \pm 45\%$, respectively). NRM intensities are fairly consistent and are of the order of 10^{-5} to 10^{-4} Am^2 kg^{-1}. All NRM directions plot in the northeastern quadrant of the equal-area plot.

The mean Koenigsberger ratio (Q_n) for the NB1 site (1.40, $N = 12$) is in the same range as the NB2 site values (1.53 ± 0.15, $N = 27$). For the SB2 site, results change with distance of the sample from the dyke margin, with lower values (1.03 ± 0.17, $N = 19$) within 1.2 m of the boundary and higher values elsewhere (1.62 ± 0.18, $N = 23$), values which are similar to their NB2 counterparts. As long as the proportion of superparamagnetic grains is almost negligible in these rocks [9], these lower Q_n ratios, which are associated with lower J_{NRM} and higher $v\%$ (Fig. 2), seem to indicate that, in this case, the magnetic grain size is larger within 1 m of the dyke boundary than elsewhere. This may be explained by greater development of high-temperature oxidation of titanomagnetic grains in the interior of the dyke, as has already been observed [24].

Strong field thermomagnetic experiments were performed only on the second set of samples, for a paleointensity study, and these have been presented elsewhere [9]. To summarise the salient details here, however, they indicate the presence of a single ferrimagnetic phase with a Curie temperature of 556 ± 10°C. The grain size of the

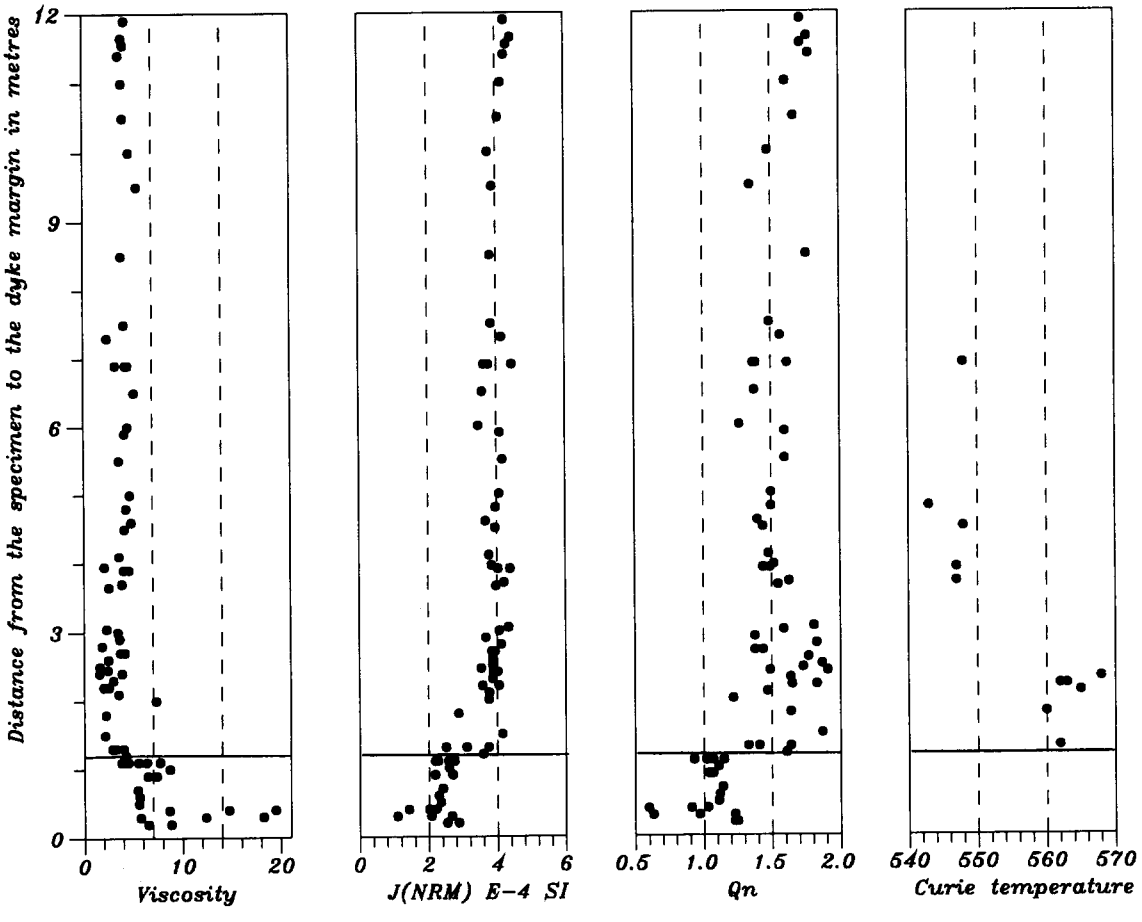


Fig. 2. Evolution of the magnetic parameters for the NB2 and SB2 samples as a function of distance of the specimens from the dyke margin. NB2 samples belong to the 2–12 m interval, SB2 samples to the 0–7.3 m interval.

magnetic minerals is of the order of 1–5 μm. The magnetic carrier of the Brenterc'h samples is therefore non-maghemitized, low-titanium pseudo-single-domain titanomagnetite.

5. Paleomagnetic results

In this section, we will present the paleodirectional analysis for the two sets of samples. The results from the first set (6 sites, 56 samples) have been presented in [1] and detailed further in [6]. In these publications, the characteristic component of magnetization per sample was defined using the best demagnetization step or the best vector difference. To ensure the coherence of the interpretation, demagnetization data from this first set of samples were reanalysed and the component of magnetization per sample defined by principal component analysis (PCA) [25]. The reader should not be surprised, therefore, if the mean directions given here are slightly different from those published previously. Site averages were calculated using Fisherian statistics [26]. No tectonic correction needs be applied (see section 3).

5.1 Brenterc'h sites

The results obtained from this site were decisive for the interpretation and therefore deserve

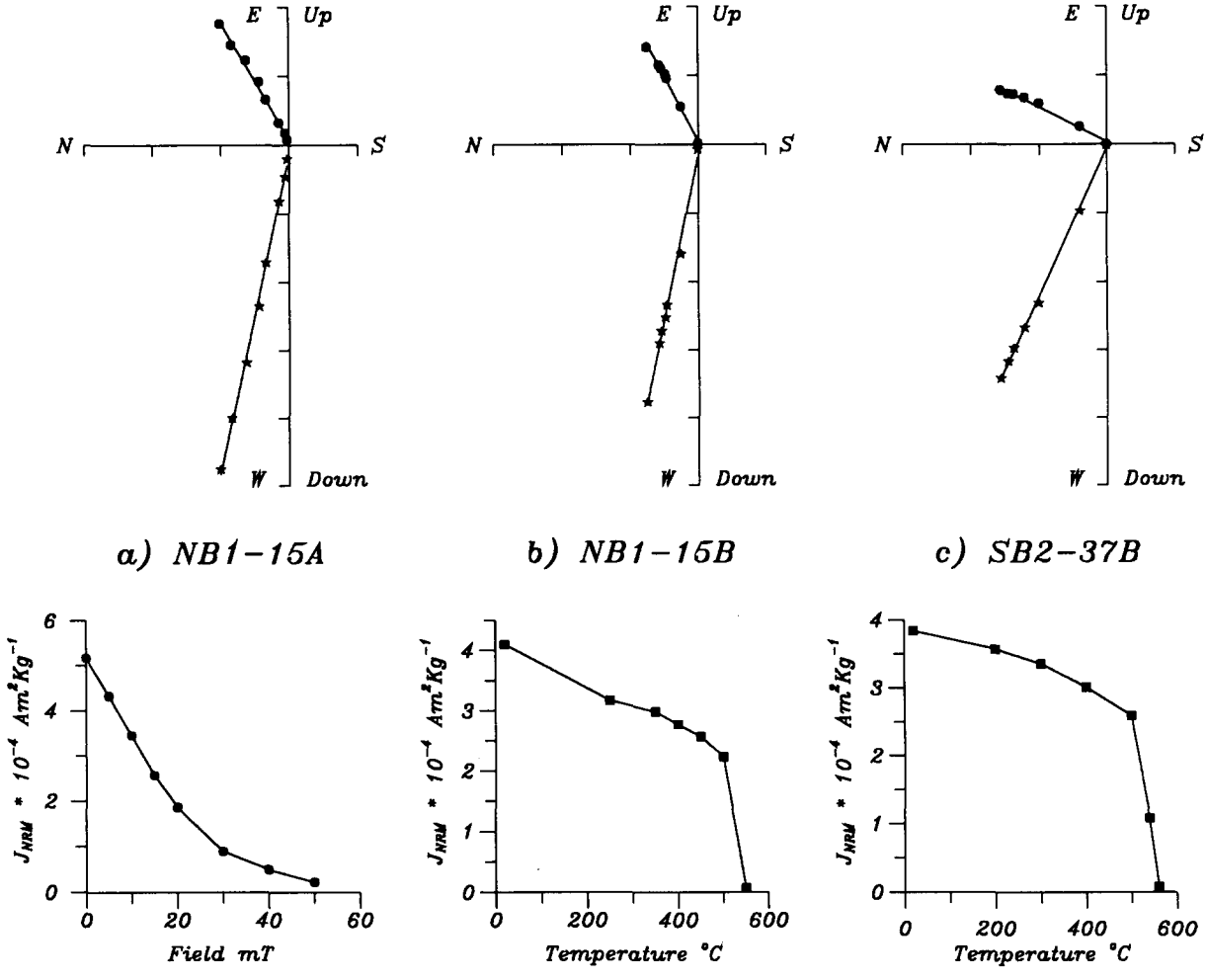


Fig. 3. Orthogonal vector diagrams (*top*) and demagnetization curves (*bottom*) for samples from the Brenterc'h site. •on the vector diagrams corresponds to the projection onto the horizontal plane; ⋆for the vertical plane. The scale for the vector diagrams is 10^{-4} Am^2 kg^{-1}.

detailed explanation. We will start with the first results obtained at Ifremer and then see what new information derives from the new results.

First set of samples: Two specimens of each sample were studied: For the NB1 specimens, one specimen was studied using AF while the other was studied using thermal treatments; all the SB1 specimens were studied using AF demagnetization. Analysis of all samples was very simple, with only one component of magnetization, although there was sometimes for a very small viscous component removed by 10–15 mT/250°C. The characteristic component of magnetization per sample was defined by PCA using between 4 and 9 demagnetization steps. The lower limit of the interval considered for the analysis is between NRM and 30 mT/350°C, the upper limit is the origin of the orthogonal projection. The PCA fits are very good, with an average maximum angular deviation (*MAD*) of 1.0 ± 0.5.

Figures 3a and b show that the components of magnetization obtained by AF and by thermal demagnetization are identical. This result can also be seen in Fig. 4, where the directions of the characteristic components of magnetization defined by the two methods [circles for AF demagnetization ($N = 7$; $I = 69.5°$; $D = 47.0°$; $k = 251$; $\alpha_{95} = 3.8°$), diamonds for heating ($N = 7$; $I = 68.4°$; $D = 46.3°$; $k = 264$; $\alpha_{95} = 3.7°$) are presented with their associated means. The mean directions associated with the two treatments and that associated with the SB1 samples (Table 2) are very well defined, emphasising the fact that secular variation has not been averaged out from this calculation and therefore that the magnetization was acquired rapidly, which indeed is the case if a thermal remanent magnetization (TRM) is considered to be the magnetization acquisition process for these samples.

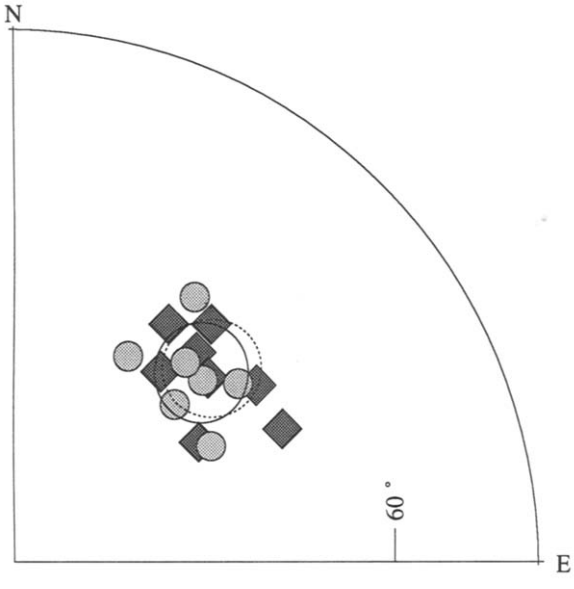


Fig. 4. Northeastern quadrant of the equal-area projection showing the clean remanence directions obtained from alternating field (AF) (shaded circles) and thermal (diamonds) demagnetizations for NB1 samples. The site-mean directions are represented with their 95% confidence limits (dashed line for thermal treatments and continuous line for AF).

TABLE 2

Paleodirectional data for Brenterc'h sites

	Site	N	H/T	I (°)	D (°)	k	α_{95}
B_3^*	NB1	7	H–T	69.5	46.9	233	4.0
B_3^*	SB1	8	H	66.7	40.0	325	3.1
B_3^*	B_3]6–9 m]	5	T	67.9	46.7	304	4.4
	B_1]0–3 m]	18	T	63.5	26.2	415	1.7
	B_2]3–6 m]	12	T	63.3	38.3	149	3.6
	B_3^*]6–9 m]	20	H–T	68.0	44.0	272	2.0
	B_4]9–12 m]	7	H–T	72.1	71.8	193	4.4
	Mean	4	H–T	67.5	42.6	97	9.4

N = number of samples; I and D = inclination and declination of the mean cleaned remanence direction; k and α_{95} = precision parameter and confidence cone about average direction. In this and in Table 3 the H/T abbreviation indicates the type of demagnetization technique used: H = AF; T = thermal.

B_3^*]6–9 m] data are for the average direction for the]6–9 m] interval using NB1 and SB1 samples and samples from the NB2 and SB2 sites sampled within this interval. The]$x-y$] notation (where $x < y$) indicates the elimination of x from the interval [$x-y$].

Second set of samples: The Brenterc'h site was reinvestigated (69 samples) for a paleointensity study. For this new problem, we were interested in the location of the samples from the dyke margin, grain size being a fairly important parameter for paleointensity studies. One specimen of each sample was thermally demagnetized prior to the paleointensity experiments as part of the preliminary experiment in order to choose the most suitable samples [9]. The characteristics of these magnetizations are similar to those of the previous sampling. Therefore, we shall only present one example of demagnetization (Fig. 3c).

The interesting point in this new study comes from the distribution of the characteristic components of magnetization calculated by PCA for each sample. Figure 5 shows a very clear relationship between the direction of magnetization, mainly declination, and the distance of the sample from the dyke margin. As long as there is no evidence for differential rotation of the dyke, the best explanation for this relationship is to consider that the samples acquired their magnetization while cooling, therefore recording secular variation of the ancient geomagnetic field. On the basis of the distribution of declinations, we divided the dyke into 3 m wide intervals and calculated mean directions per interval (Table 2, Fig. 6) as best estimates of instantaneous recording of the field. These mean directions are equivalent to the mean directions which will be defined for the other sites sampled along the dyke. The mean directions associated with the NB1 and SB1 sites indicate that their samples have been sampled essentially within the]6–9 m] interval (Fig. 6). In order to avoid giving too much weight to this interval in the calculation of the overall mean direction, the NB1 and SB1 results were included in the calculation of the mean direction associated with the]6–9 m] interval (B_3^*,]6–9 m], Table 2).

Averaging the poles associated with the four intervals gives a dispersion parameter K of 38. This dispersion parameter can be related to an angular scatter of 13.5°, which is rather small compared to the minimum VGP angular scatter, which is estimated at 16.5° from paleosecular variation studies of lava flows (PsVL) [27] at this latitude for the 110–195 Ma period. This could

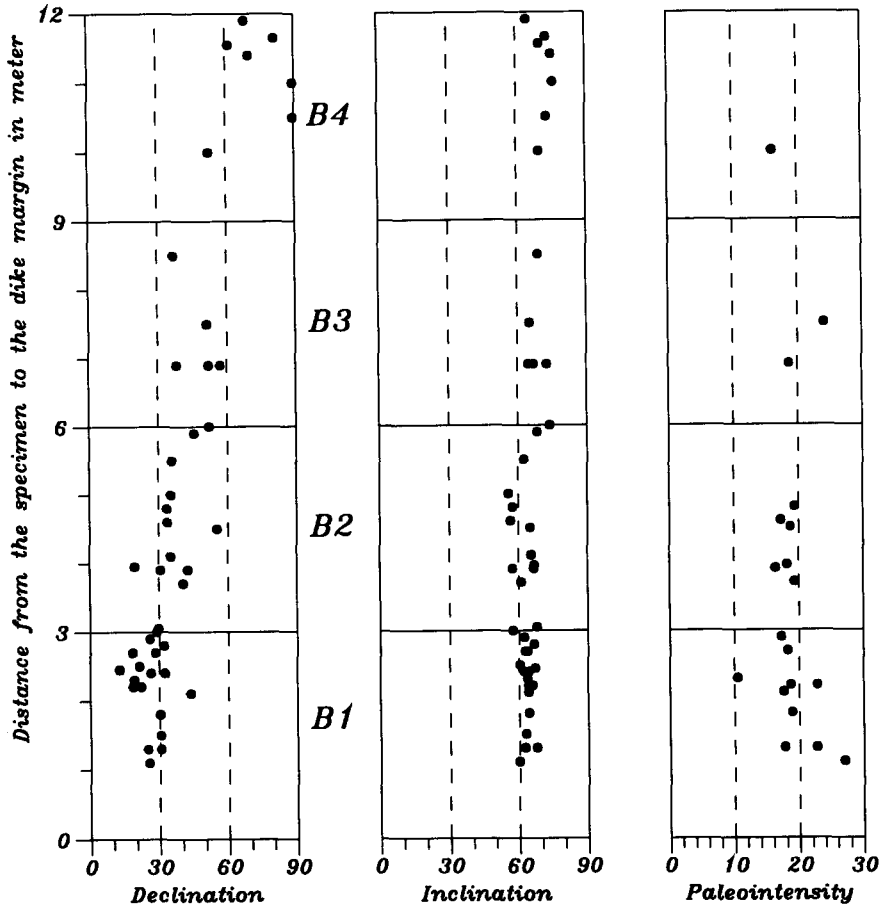


Fig. 5. Evolution of the clean remanence directions and paleointensity (μT) estimates for the NB2 and SB2 samples as a function of the distance of the specimens from the dyke margin.

indicate that only part of the paleosecular variation was averaged out at this site. A model of instantaneous cooling of a 30 m wide dyke embedded in gneiss indicates a difference of only 10 yrs between cooling, both at the centre of the dyke and at the its margins, under the 575°C isotherm. However, such a model would mean that a shift of the pole of 19° (angular difference between B_1 and B_4, Table 2) would occur in 10 yrs (i.e., at 1.9°/yr). This value is up to ten times higher than the generally assumed rate of 0.1–0.2°/yr, and at this rate the cooling time would be of the order of 100–200 yrs. This crude estimation demonstrates the principal that simple models are far removed from nature, and the setting of the Brenterc'h dyke was not instantaneous. This result is not surprising when compared to setting time estimates of up to a million years for comparable extension dyke systems [Dautria, pers. commun.].

These results are crucial because they confirm the reality of the acquisition of the magnetization by TRM and therefore the primary character of the magnetization. This is, of course, important for the paleointensity study but also for the paleodirectional analysis and underline the fact that this direction can no longer be discarded as not reliable.

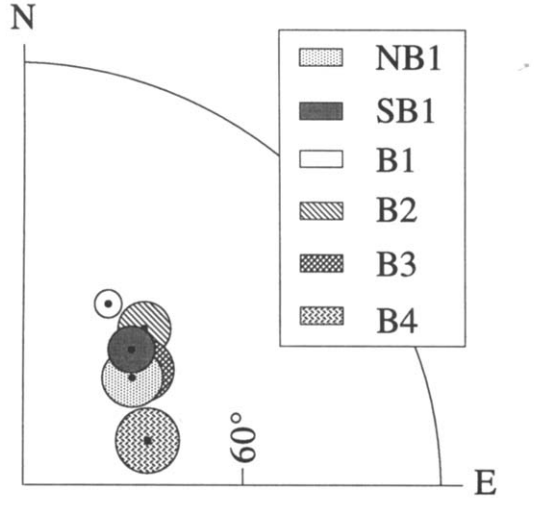


Fig. 6. Northeastern quadrant of the equal-area projection showing the site-mean clean remanence directions for the Brenterc'h sites.

5.2 Other sites

Four other sites were studied [1,6], mainly by stepwise AF demagnetization. Principal component analysis was conducted between NRM and 30 mT and the origin of the orthogonal diagrams, resulting in between 4 and 9 demagnetization steps.

Samples from the Porsmilin site (PN) have fairly low viscosity indexes and show only one component of magnetization. Surprisingly, however, four samples were completely remagnetized in the present field. The other three samples carry a characteristic component of magnetization similar to that at the other sites (Figs. 7a and 8).

The mean viscosity index of samples from the Douarnenez site (DZ) is higher than for the previously quoted sites (NB1, SB1 and PN). This property is reflected by a strong decrease in the intensity of magnetization between NRM and 5 mT (Fig. 7b). After the removal of this viscous component the characteristic components of magnetization can be defined (Figs. 7b and 8) with good precision (*MAD* of the order of 1.0 ± 0.7).

In the case of samples from the Quimper site (QR), the viscosity index is very large. The decrease in intensity of magnetization between NRM and 5 mT is very important (Fig. 7c) and the associated vector differences are very scattered, which can mean that the viscous components were acquired while the samples were awaiting treatment in the laboratory. After removal of these viscous components, only five of the eight samples allowed calculation of a stable component of magnetization (Figs. 7c and 8), and the associated errors were much larger ($MAD = 2.7 \pm 1.1$). For the other three samples, we found it necessary to use the last reliable demagnetization step to approach clean remanence. AF determinations were used in the calculation of the mean clean remanent direction.

The last samples come from the Quemenes site (QS). The magnetization here is very unstable, and this is certainly because of weathering at the outcrop. Furthermore, it was noticed that some samples caught spurious components of magnetization while entering the mu-metal shields of the magnetometer. Demagnetization of

these samples was therefore ended and Quemenes was omitted from any further analysis.

5.3 Mean results

The mean results for the complete set of samples from the Kerforne dyke are summarized in Table 3 and presented in Fig. 8. Considering the number of samples and the relatively small geographical extent for each site (with the exception of Brenterc'h) it is clear that secular variation is not averaged out site by site. Therefore, we gave the same weight to the mean directions associated with these sites and to those calculated per interval at Brenterc'h. We then defined a mean direction associated with the lower Jurassic Kerforne dyke (Table 3) with $D = 41°$ and $I = 62°$ ($k = 66$, $\alpha_{95} = 7.5°$) and a mean paleopole having a latitude of 61°N and a longitude of 79°E ($K = 36$, $A_{95} = 10.2°$).

Here again the angular scatter of 13.5° which can be computed from the seven sites is smaller than the 16.5° angular scatter expected from paleosecular variation studies of lava flows PSVL [27] at this latitude for the 110–195 Ma period. Nevertheless it is worth recalling that the intrinsic uncertainty, at the 95% confidence limit, in the determination of K (which at $\pm 20\%$ is always high, even with 100 determinations [29]), is extremely large for seven sites. This precludes any

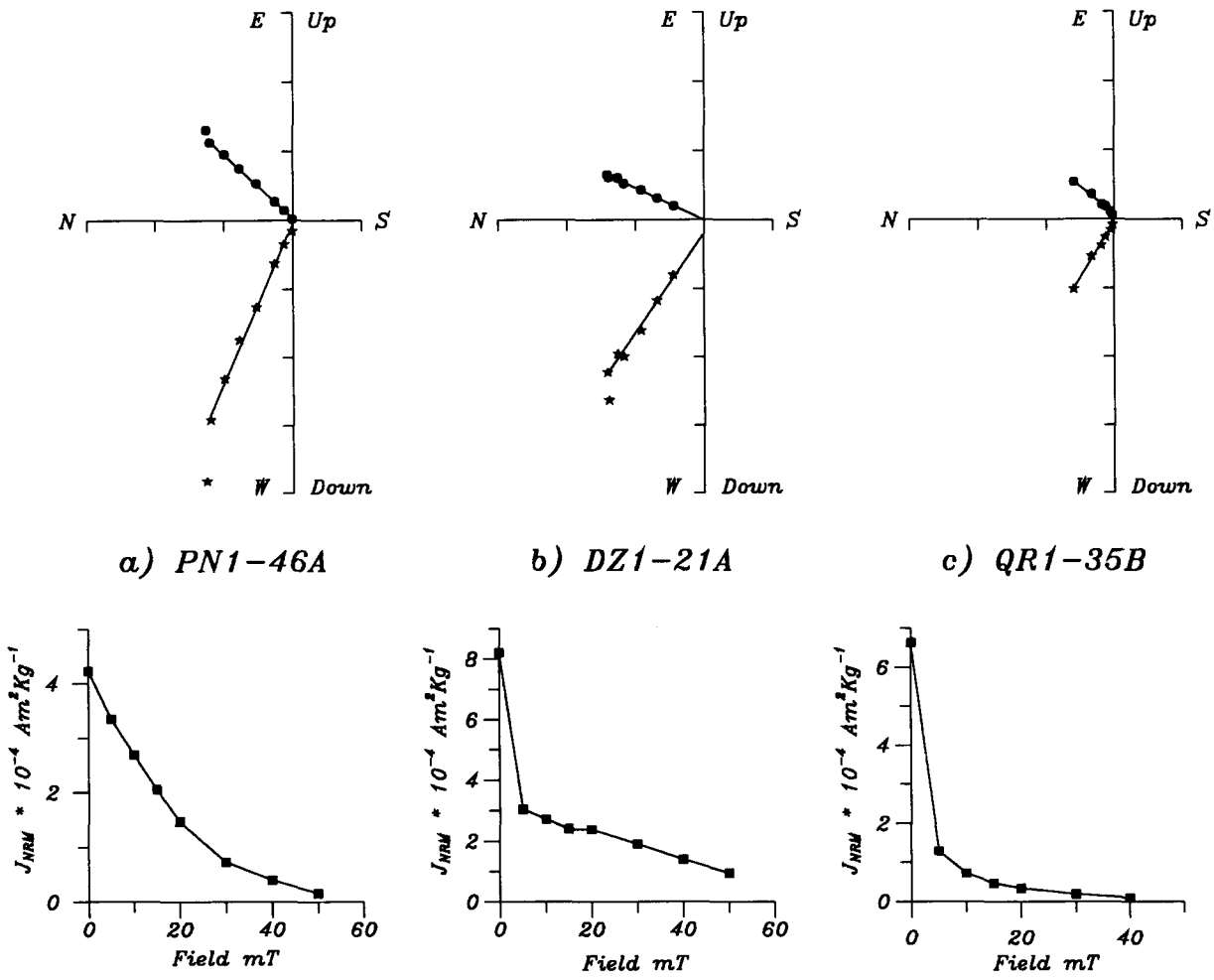


Fig. 7. Orthogonal vector diagrams (*top*) and demagnetization curves (*bottom*) for samples from some other sites along the Kerforne dyke. To enlarge the centre part of the orthogonal diagrams for samples DZ1-21A and QR1-35B, the NRM points were omitted. See Fig. 3 for further explanation.

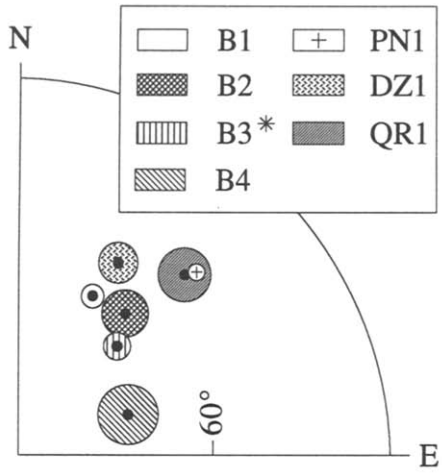


Fig. 8. Northeastern quadrant of the equal-area projection showing the site-mean clean remanence directions for the Kerforne sites.

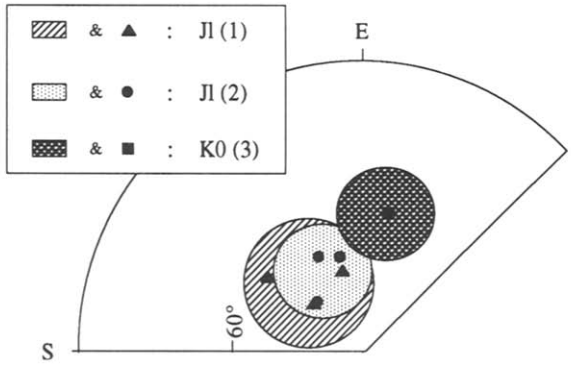


Fig. 9. Comparison of the Kerforne VGP with selected European lower Jurassic paleopoles and associated means. (1) Selection by Van der Voo [31] and (2) by Besse and Courtillot [32]. (3) Kerforne pole.

further estimation of the true uncertainty associated with this pole, which may be more than the 10° reflected by the A_{95}.

6. Discussion

The Virtual Geomagnetic Pole yielded by the paleomagnetic study of the Kerforne dolerites is notably off the mean lower Jurassic paleopoles computed from European data by several other authors using different selection criteria and smoothing methods [e.g., 5,8,30,31,32]. When compared with two very recent compilations [31,32] the angular difference is, respectively, 22° and 18° (Fig. 9) and there is almost no overlap between the confidence circles.

The first way of explaining this discrepancy is to call into question the age and/or the magnetization of the Kerforne dolerites.

The numerous radiometric results obtained from the dyke and listed at the beginning of this paper indicate an age of between 190 and 200 Ma, which is in good agreement with ages from other circum-Atlantic dolerites [33,34,35] which are also understood to be related to the first rifting episodes of the Atlantic Ocean. Moreover, a bias in age would not be capable of explaining the observed discrepancy in the Kerforne VGP position.

A bias in the mean component of magnetization could be due to secondary components that were not cancelled. As may be seen from the analysis of the Kerforne samples, the direction is essentially univectorial. The magnetization was

TABLE 3

Mean paleodirectional results from the lower Jurassic Kerforne dyke

Site	N	H/T	I (°)	D (°)	k	α_{95}	PLat	PLong	K	A_{95}
B_1	18	T	63.5	26.2	415	1.7	71.9	85.5	196	2.5
B_2	12	T	63.3	38.3	149	3.6	63.7	77.2	80	4.9
B_3^*	20	H-T	68.0	44.0	272	2.0	61.9	62.5	110	3.1
B_4	7	H-T	72.1	71.8	193	4.4	47.7	44.9	72	7.2
PN1	3	H-T	51.0	45.2	81	(13.8)	52.6	94.1	70	(14.8)
DZ1	8	H	57.4	28.9	352	3.0	66.9	100.6	186	4.1
QR1	8	H-T	52.7	43.6	199	3.9	54.5	93.1	121	5.1
Mean	7	H-T	61.7	41.0	66	7.5	61.3	78.8	36	10.2

N = number of points in the average calculation; 'PLat' and 'PLong' = north latitude and east longitude of the Virtual Geomagnetic Pole; K and A_{95} = precision parameter and confidence cone about average pole. See Table 2 for further explanation.

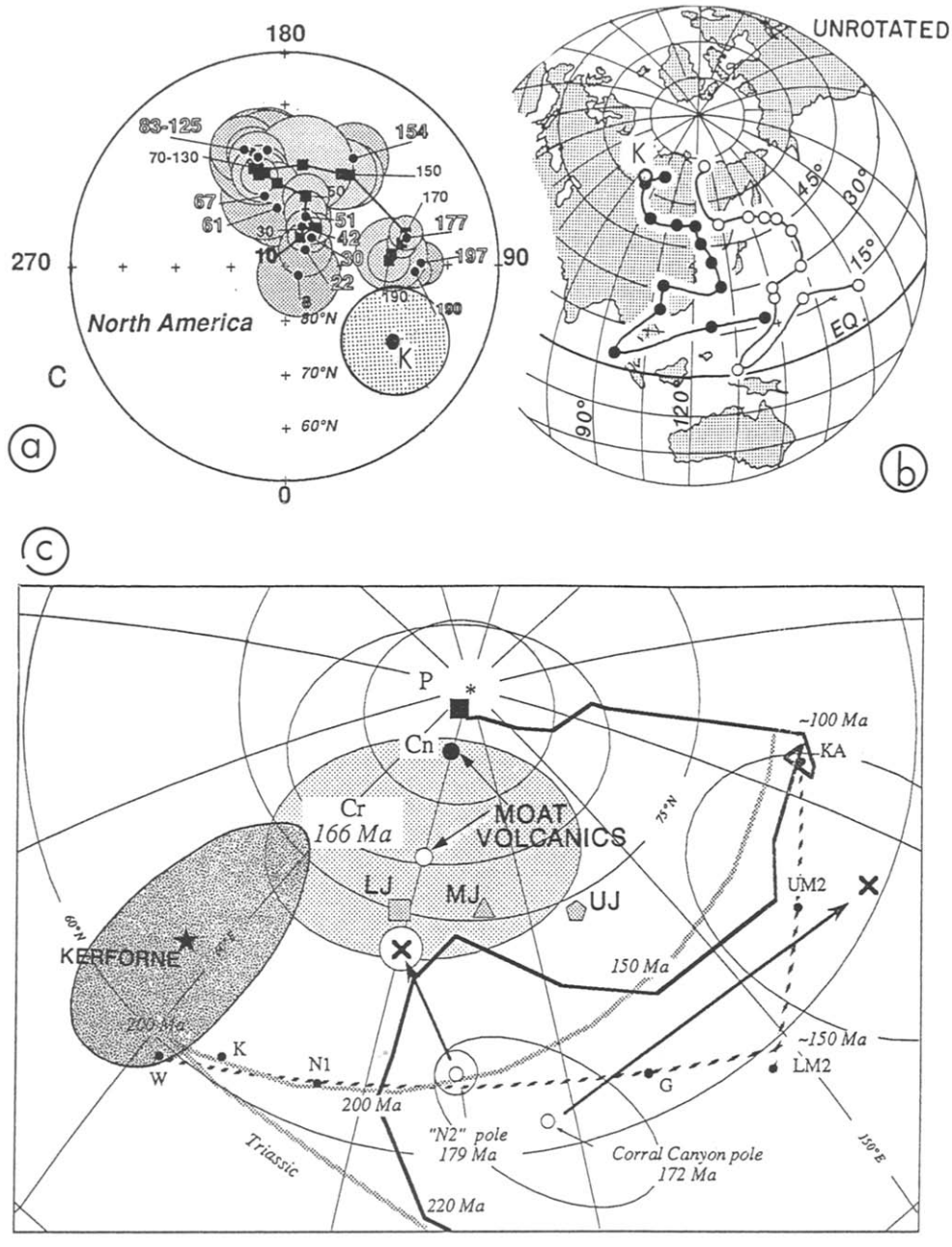


Fig. 10. Comparison of the Kerforne VGP with selected North American APWPs. (a) Van der Voo [31]. (b) Besse and Courtillot [32]. (c) Van Fossen and Kent [38]. The Kerforne pole was transfered into North American coordinates using Olivet et al's. [44] rotation parameters for (a) and (c). Kerforne pole in present-day coordinates for (b). Open symbols for the European APWPs, closed symbols for the American APWPs.

acquired rapidly by TRM processes and only part of the secular variation may have been cancelled out. Further, it may be recalled that, in a study on the Messejana dolerite dyke on the Iberian peninsula, which is roughly estimated to be the same age as the Kerforne dyke [28], the angular scatter between the twelve sites is exactly the same (13.5°) with regard to the same minimum

VGP angular scatter, and in this study one can assume that the paleosecular variation is probably quite well cancelled. Therefore, this second hypothesis can reasonably be eliminated as a major cause of the discrepancy.

If we accept the reliability of the Kerforne VGP, acknowledging nevertheless the fact that it may still have a bias related to a not perfectly cancelled paleosecular variation, the proposed European APWPs for the Triassic–Jurassic can then be called into question. Considering the small number of reliable lower Jurassic data available for Europe, the Kerforne pole was therefore compared with North American results. Selected North American APWPs are plotted in Fig. 10 with the Kerforne pole after rotation to North American coordinates (closure of the Atlantic Ocean) using Bullard et als.' parameters [35], which are those preferred in [31]. The differences in data selection associated with the use, or not, of smoothing methods led to fairly different APWPs, as has already been pointed out by several authors [e.g., 36,37]. Besse and Courtillot's [32] compilation listed only paleomagnetic results that fulfilled their minimum reliability criteria. The APWP from the present back to 200 Ma reproduced in Fig. 10c was obtained by connecting mean poles averaged every 10 Ma with a 20 Ma sliding window. The Van der Voo [31] compilation listed all avalaible paleomagnetic results fully based on demagnetization techniques with an associated quality index (Q). The APWP for the middle Ordovician–early Jurassic interval reproduced in Fig. 10b has been constructed by connecting mean poles calculated using results having a Q of $\geqslant 3$ averaged for periods of about

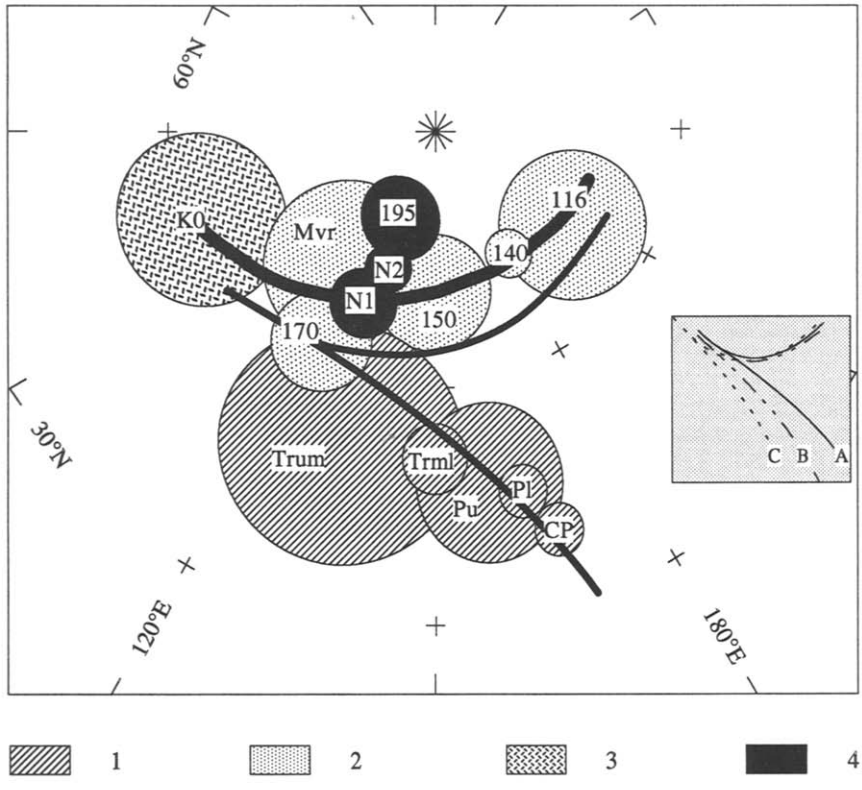


Fig. 11. Synthesis of European APWPs from the Permo-Carboniferous through to the upper Jurassic. *1* = Mean *C–P–Tr* VGPs [31]; *2* = Jurassic and early Cretaceous VGPs [32]; *3* = Kerforne VGP; *4* = high-latitude Jurassic poles for North America in European coordinates. Very thick line = European Jurassic–Cretaceous APWP computed by PEP analysis; less thick lines = North American late Carboniferous–Jurassic APWP [42] computed by PEP analysis in European coordinates. The North American poles and path were rotated using Bullard et al's, fit [35]. The inset shows different positions of the North American APWP depending on the rotation parameters used (*A*—[35]; *B*—[43]; *C*—[44]).

25 m.y. with no sliding average. Finally Van Fossen and Kent's analysis [38], which is presently at the centre of a debate [39,40], is also reproduced here (Fig. 10c). In addition, there are some other APWPs reported in this paper: Irving and Irving's [30] APWP (heavy black line in Fig. 10c), which is also based on a running average and has since been updated, is one but this will not be considered any further here. May and Butler's [37] APWP (dashed line in Fig. 10c) is based on rigorous data selection with no averaging. This is also true for Gordon et al's. selection, but they defined their APWP by Paleomagnetic Euler Pole (PEP) analysis [41]. Following this concept, APWP are considered as long, gently curved segments, termed 'tracks', linked by short segments with sharp curvatures, termed 'cusps' or 'hairpins'. Therefore, the most westerly North American points shown in Fig. 10c should not be averaged with the other data because they represent the late Triassic–early Jurassic cusp [37,41] related to a change in North American absolute plate motion associated with rifting of the Central Atlantic. We can see that, albeit with an almost 20° difference between the Kerforne pole in North American coordinates and the mean lower Jurassic poles computed by Van der Voo [31] (Fig. 10a) and Besse and Courtillot [32] (Fig. 10b), the agreement is much better with the two APWPs [37,41] (which use no averaging), the Kerforne pole falling close to the lower Jurassic cusp. The result is similar to that obtained from mean European data. Could the Kerforne pole represent the first indication of a very westerly cusp in the European data?

In order to prove this hypothesis, it would be necessary to conduct a PEP analysis based solely on European Jurassic data, and unfortunately such data are still very scarce (only six reliable data for the 195–140 Ma interval [32]). Even so, we have estimated the small circle (heaviest line in Fig. 11) which best fits early Jurassic–early Cretaceous European data to facilitate comparison with updated PEP *C–P–Tr* and *J–K* tracks proposed for North America [42]. Different positions for the tracks are shown (inset in Fig. 11), depending on the considered rotation parameters (*A*, *B* and *C* in [35], [43] and [44] respectively). These three sets of parameters were chosen because they are representative of the maximum range of variation in the track positions; most other proposed rotations will make the tracks fall in between curves *A* and *C*. The choice of the rotation parameters is especially important for the *C–P–Tr* track but has almost no influence on the *J–K* track. We finally selected Bullard et al's. continental reconstruction [35], which was shown to best fit the paleomagnetic data [31]. If the rotated *C–P–Tr* track well fits mean European paleopoles computed for the Carboniferous–Triassic period [31] (ornament *1*, Fig. 11), the European Jurassic data (ornament *2*, Fig. 11) are systematically offset from the rotated *J–K* track towards higher latitudes. A controversy already exists regarding the Jurassic APWP for North America [38] and the reliability of high-latitude paleopoles, some of them being interpreted as overprints. Three of these poles are also plotted, after rotation in European coordinates (ornament *4*, Fig. 11: N1, Newark Supergroup volcanics [45]; N2, Newark Basin overprint [46]; Mvr, Moat volcanics reversed overprint [38]). The White Mountains P component and the Moat volcanics normal polarity secondary component [38] are not considered here because it is not clear from the presented data whether these two components are truely free of a residual present-day field component. While the Newark poles are in very good agreement with the European data, the Mvr pole is still higher in latitude. This seems to support the reality of a higher latitude track in Jurassic time, although considering the associated errors this is not conclusive. Nevertheless, in this framework the Kerforne pole (ornament *3*, Fig. 11) seems to be the first indication of the existence of a cusp around 200 Ma in the European APWP.

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